

Training our intuition: two threads

- How many interleavings if we have a thread $A = \{A1, A2\}$ and a thread $B = \{B1\}$?

Training our intuition: two threads

- How many interleavings if we have a thread $A = \{A1, A2\}$ and a thread $B = \{B1\}$?
- *Reminder:* in the concurrency abstraction, we don't reorder actions within threads
 - so $A2$ always follows $A1$
 - other than that, we mix atomic actions in all possible ways
- *Answer:* 3
 - the interleavings are $[A1, A2, B1]$, $[A1, B1, A2]$, $[B1, A1, A2]$

Training our intuition: two threads (variant)

- How many interleavings if we have a thread $A = \{A1, A2\}$ and a thread $B = \{B1, B2\}$?

Training our intuition: two threads (variant)

- How many interleavings if we have a thread $A = \{A1, A2\}$ and a thread $B = \{B1, B2\}$?
- Let's start enumerating
 - if A1 is first: $[A1, A2, B1, B2]$, $[A1, B1, A2, B2]$, $[A1, B1, B2, A2]$
 - If B1 is first: $[B1, A1, A2, B2]$, $[B1, A1, B2, A2]$, $[B1, B2, A1, A2]$
 - no interleaving starts with A2 or B2
- *Answer: 6*
 - note: one more action $\rightarrow$ double the number of interleavings...

Training our intuition: any two threads

- Can you derive a general formula of the number of interleavings for two threads A and B if A has X atomic actions and B has Y actions?

Training our intuition: any two threads

- Can you derive a general formula of the number of interleavings for two threads A and B if A has X atomic actions and B has Y actions?

- This is basically a *stars-and-bars* calculation



- if we have n stars and m bars, the number of possible sequences (including those with consecutive bars) is the binomial coefficient of $n+m$ over n
- *Answer:* $(X+Y)! / X!Y!$

Training our intuition: three threads

- How many interleavings if thread $A = \{A1\}$, thread $B = \{B1\}$ and thread $C = \{C1\}$?

Training our intuition: three threads

- How many interleavings if thread $A = \{A1\}$, thread $B = \{B1\}$ and thread $C = \{C1\}$?
- Let's enumerate them
 - A1 first: $[A1, B1, C1]$, $[A1, C1, B1]$
 - B1 first: $[B1, A1, C1]$, $[B1, C1, A1]$
 - C1 first: $[C1, B1, A1]$, $[C1, A1, B1]$
- *Solution: 6*

Training our intuition: three threads (variant)

- How many interleavings if thread $A = \{A1, A2\}$, thread $B = \{B1\}$ and thread $C = \{C1\}$?

Training our intuition: three threads (variant)

- How many interleavings if thread $A = \{A1, A2\}$, thread $B = \{B1\}$ and thread $C = \{C1\}$?
- Let's start enumerating
 - A1 first: $[A1, A2, B1, C1]$, $[A1, A2, C1, B1]$, $[A1, B1, A2, C1]$, $[A1, B1, C1, A2]$, $[A1, C1, A2, B1]$, $[A1, C1, B1, A2]$
 - B1 first: $[B1, A1, A2, C1]$, $[B1, A1, C1, A2]$, $[B1, C1, A1, A2]$
 - C1 first: $[C1, B1, A1, A2]$, $[C1, A1, B1, A2]$, $[C1, A1, A2, B1]$
- *Solution:* 12
 - twice than if we had 2 threads with 2 actions each...

Training our intuition: simple program

- What are the possible values of x after the termination of $T1: \{x=5; x=2*x\}$ and $T2: \{x=x+2\}$?

Training our intuition: simple program

- What are the possible values of x after the termination of $T1: \{x=5; x=2*x\}$ and $T2: \{x=x+2\}$?
- Let's apply the same methodology as before
 - interleaving1: $x=5; x=2*x; x=x+2; \rightarrow x=12$
 - interleaving2: $x=5; x=x+2; x=2*x; \rightarrow x=14$
 - interleaving3: $x=x+2; x=5; x=2*x; \rightarrow x=10$

Training our intuition: simple program

- What are the possible values of x after the termination of $T1: \{x=5; x=2*x\}$ and $T2: \{x=x+2\}$?
- Let's apply the same methodology as before
 - interleaving1: $x=5; x=2*x; x=x+2; \rightarrow x=12$
 - interleaving2: $x=5; x=x+2; x=2*x; \rightarrow x=14$
 - interleaving3: $x=x+2; x=5; x=2*x; \rightarrow x=10$
- But are all those instructions atomic?
 - $x=2*x$ entails read & write: can be written as $t=x; x=2*t;$
 - same for $x=x+2$, which can be written as $s=x; x=s+2;$

Training our intuition: simple program

- What are the possible values of x after the termination of $T1: \{x=5; x=2*x\}$ and $T2: \{x=x+2\}$?
- But are all those instructions atomic?
 - $x=2*x$ entails read & write: can be written as $t=x; x=2*t;$
 - same for $x=x+2$, which can be written as $s=x; x=s+2;$
- So additional values are possible!
 - interleaving4: $s=0; T1; x=s+2; \rightarrow x=2$
 - interleaving5: $s=0; x=5; x=s+2; x=2*x; \rightarrow x=4$
 - interleaving6: $x=5; s=5; ...; x=s+2 \rightarrow x=7$